6 Baseball Elimination

Suppose T is a set of teams in a sports league. At any point during the season, each time i has some number of wins, w_i and some number of games left to play g_i (for each team j, team i has $g_{i,j}$ games left against j). Our goal is to determine if team i still has a chance to finish in first place. Note that sometimes a team cannot do this, even if they win every remaining game. For purposes of this problem, games cannot end in a tie.

Let $W_k = w_k + g_k$ be the maximum number of wins possible for team k. There are two ways for team k to be eliminated:

- $W_k < w_i$ for some team i. That is, even if k wins out and i loses out, i still finishes with more wins.
- Some combination of games "mathematically eliminate" team k; that is, there is a set of teams T that will finish with a *strictly higher* average number of wins than W. Note that this means the league does not allow games to end in a tie. More formally:

$$\sum_{x \in T} w_x + \sum_{x,y \in T} g_{x,y} > W \cdot |T|$$

We are going to use network flow techniques to determine if a given team k has been eliminated. The idea is going to be to use network flow to determine if there is a way to allocate wins across every series such that no team has strictly more wins than team k. The resulting flow should allow us to either determine that team k has a possible outcome in which they finish with the most wins, or can demonstrate to an observer (who hasn't taken this class) why that cannot happen.

Let's see a few examples.

Question 16. It is near the end of the season and teams A, B, C, and D are each in contention. Team A has 92 wins, B and C have 91 wins each, and D has 90 wins. All pairs of those teams have one game remaining, *except* A and D will not play one another. Can team D finish with at least a tie for first?

Question 17. Consider the following table of games remaining. Has team D been eliminated?

Team	Wins	Games Left	Schedule ($g_{i,j}$)			
i	w_i	g_i	A	B	C	D
A	90	11		1	6	4
B	88	6	1		1	4
C	87	11	6	1		4
D	79	12	4	4	4	

Question 18. Use network flow to solve the general baseball elimination problem

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Activity: Sticker Problem

Professor Shindler has n students taking CompSci 161 this quarter and m different types of stickers he wants to give to students. He wants to only give students stickers that they will like. He wants to give away all the stickers but recognizes that not everyone will receive one (that is, the total number of stickers is less than n). Each student will receive either 0 or 1 stickers.

- Each student has a wish list of stickers; that is, for each student, we know which subset of the m types of stickers that student wants to receive.
- For each sticker type i, he knows that he has r_i copies of the sticker available.

Use network flow techniques to give a polynomial time algorithm to decide if it is possible to distribute all the stickers with these constraints. If it is possible, describe how you can determine which student receives which type of sticker.

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